

# Collaborative Ocular Melanoma Study (COMS) Randomized Trial of I-125 Brachytherapy for Medium Choroidal Melanoma

## I. Visual Acuity after 3 Years COMS Report No. 16

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Collaborative Ocular Melanoma Study Group

**Objective:** To report visual acuity during the first three years after iodine 125 ( $I^{125}$ ) brachytherapy for medium-sized choroidal melanoma and to identify important baseline and treatment factors associated with posttreatment visual acuity in a group of patients who were treated and observed prospectively as part of a large, randomized clinical trial.

**Design:** Observational case series within a randomized, multicenter study.

**Participants:** Patients enrolled in the Collaborative Ocular Melanoma Study randomized trial of  $I^{125}$  brachytherapy versus enucleation had choroidal melanoma of at least 2.5 mm but no more than 10.0 mm in apical height, and no more than 16.0 mm in largest basal dimension. One thousand three hundred seventeen patients enrolled from February 1987 through July 1998; 657 patients were assigned to  $I^{125}$  brachytherapy. Visual acuity data for 623 patients who received  $I^{125}$  brachytherapy as randomly assigned and who have been observed for at least 1 year were analyzed for this report.

**Methods:** Under study protocol, an ophthalmic evaluation, including best-corrected visual acuity measurement of each eye, was performed at baseline, every 6 months thereafter for 5 years, and once yearly thereafter. Two poor vision outcomes, visual acuity of 20/200 or worse that was confirmed at the next follow-up examination and loss of six lines or more of visual acuity from baseline that was confirmed at the next follow-up examination, were analyzed to identify baseline and treatment characteristics that were associated with posttreatment visual acuity.

**Results:** At baseline, median visual acuity in the eye with choroidal melanoma was 20/32, with 70% of eyes having 20/40 or better and 10% of eyes having 20/200 or worse visual acuity. Three years after  $I^{125}$  brachytherapy, median visual acuity was 20/125, with 34% having 20/40 or better and 45% having 20/200 or worse visual acuity, including eyes that were enucleated within 3 years of treatment. Life-table estimates of percentages of patients who lost six or more lines of visual acuity from baseline, a quadrupling of the minimum angle of resolution, with this finding confirmed at the next 6-month follow-up examination, were 18% by 1 year, 34% by 2 years, and 49% by 3 years after treatment. Life-table estimates of percentages of patients with baseline visual acuity better than 20/200 whose visual acuity decreased to 20/200 or worse, confirmed at the next follow-up examination, were 17% by 1 year, 33% by 2 years, and 43% by 3 years after treatment. As soon as a poor vision outcome was observed, improvement of visual acuity to a level that no longer met the definition for a poor vision outcome was rare. Greater baseline tumor apical height and shorter distance between the tumor and the foveal avascular zone (FAZ) were the factors most strongly associated with loss of six or more lines of visual acuity after treatment. These two factors and baseline visual acuity also were strongly associated with visual acuity 20/200 or worse after treatment. Patient history of diabetes, presence of tumor-associated retinal detachment, and tumors that were not dome shaped also were associated with greater risk for both of the poor vision outcomes.

**Conclusions:** Forty-three percent to 49% of treated eyes had substantial impairment in visual acuity by 3 years after  $I^{125}$  brachytherapy, defined as a loss of six or more lines of visual acuity from the pretreatment level (49% of eyes) or visual acuity of 20/200 or worse (43% of eyes) that was confirmed at the next 6-month examination. Patients with a history of diabetes and patients whose eyes had thicker tumors, tumors close to or beneath the FAZ, tumor-associated retinal detachment, or tumors that were not dome shaped were those most likely to have a poor visual acuity outcome within 3 years after  $I^{125}$  brachytherapy. *Ophthalmology* 2001;108:348-366 © 2001 by the American Academy of Ophthalmology.

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